Reading and Writing

33 QUESTIONS

DIRECTIONS

The questions in this section address a number of important reading and writing skills. Each question includes one or more passages, which may include a table or graph. Read each passage and question carefully, and then choose the best answer to the question based on the passage(s).

All questions in this section are multiple-choice with four answer choices. Each question has a single best answer.

1

The following text is adapted from Anton Chekhov's 1904 play The Cherry Orchard (translated by Julius West in 1916).

TROFIMOV: Believe me, Anya, believe me! I'm not thirty yet, I'm young. I'm still a student, but I have undergone a great deal! I'm as hungry as the winter, I'm ill, I'm shaken....and where haven't I been—fate has tossed me everywhere!

As used in the text, what does the word "undergone" most nearly mean?

A) Neglected
B) Enjoyed
C) Conveyed
D) Endured

2

Indigenous Photograph is an organization whose mission is to ensure that images of indigenous peoples in the media are presented from indigenous perspectives. The organization ___ this commitment by promoting the works of artist Geremew Tigabu (Ethiopian, Amhara, and Tigre) and other prominent indigenous photographers who document and reflect Indigenous lives and experiences.

Which choice completes the text with the most logical and precise word or phrase?

A) concludes
B) explains
C) precedes
D) shows

3

Although fewer companies trade their stocks on the Cambodia Securities Exchange in Phnom Penh, Cambodia, than on the stock exchanges in London, Mumbai, or Tokyo, the Cambodia Securities Exchange has the advantage of being able to ____ relatively small companies in Cambodia: by connecting those companies to investors with expertise about the country's economic conditions, the Cambodia Securities Exchange can help those companies thrive.

Which choice completes the text with the most logical and precise word or phrase?

A) designate
B) nurture
C) preclude
D) assess

4

The emphasis on accurately representing the experiences of average working people that is characteristic of the realist style can be seen in The Gleaners, painted by Jean-Francois Millet, which depicts peasants picking stray wheat from a field after the harvest. This style can thus be seen as an effort to ___ what were regarded as the excesses of the romantic style evident in many paintings by Horace Vernet, which instead exaggerated their subjects' beauty or heroism while hiding all imperfection.

Which choice completes the text with the most logical and precise word or phrase?

A) understand
B) advance
C) counteract
D) accentuate

5

The swordfish can swim very fast-up to 97 kilometers per hour (km/hr) —but it is significantly slower than the frigatebird, which can fly at speeds up to 153 km/hr. The difference between these speeds is largely____ of the fact that the features that make flight possible do less to limit top speeds than the features suitable for swimming through water.

Which choice completes the text with the most logical and precise word or phrase?

A) a consequence
B) an explanation
C) a repudiation
D) an objective

6

<u>President Richard Nixon is most famous for his participation in the 1970s Watergate political scandal, a convoluted tale of criminality and eroded ethics involving a constellation of associates such as security operative Jack Caulfield and Attorney General John Mitchell.</u> But Nixon's legacy is complex: he has been praised for his role in affirming the sovereignty of tribal nations, and he once made an attempt at reforming United States health care policy that is arguably a precursor to the Affordable Care Act, which became law during the Barack Obama administration.

Which choice best describes the function of the underlined sentence in the text as a whole?

A) It presents an accomplishment of a historical figure whose significance is detailed later in the text.
B) It describes a common perception of a historical figure that is challenged by information presented later in the text.
C) It states a claim about a historical figure that is supported by evidence later in the text.
D) It compares the achievements of three historical figures to a fourth that is mentioned later in the text.

7

In what is now Washington state, the Tulalip Tribes operate the Hibulb Cultural Center. Relying on traditional knowledge to guide the design of exhibits, this institution persents Tulalip history and culture to the tribes' citizens. The Comanche Nation, a tribe in Oklahoma, employs a similar strategy in its own cultural center. Both centers contrast with museums that aren't Indigenous-led; when displaying Indigenous artifacts, such museums tend to anticipate mainly non-Indigenous audiences and rely on Euro-centric strategies for designing exhibits.

Which choice best describes the function of the underlined sentence in the text as a whole?

A) It suggests improvements to a particular tribal cultural center.

B) It encourages tribal citizens to attend their local cultural center.

C) It explains how one tribal cultural center differs from other tribal cultural centers.

D) It provides a basic description of a particular tribal cultural center.

8

The museum of Modern Art (MOMA) in New York City has an exhibition of video games that includes Pac-Man from 1980, which museum visitors can play on site, and SimCity 2000 from 1994, which visitors can see only in a video presentation. MOMA claims the video presentations are only for games that would be impractical to display in a playable form, but video games are an inherently interactive medium, a feature that is grossly absent in a video-only presentation.

Which choice best describes the function of the underlined sentence in the text as a whole?

A) It identifies a feature of many video games that is not shared by some of the games included in MOMA's exhibition.

B) It provides a claim about video games as art that both MOMA and the author accept as true.

C) It describes a misconception about video games that the author believer is evident in MOMA's choice about which video games to exhibit.

D) It presents a consideration that the author thinks partly undermines MOMA's approach to exhibiting video games.

9

In O'odham, an Indigenous language from the Southwest region of what is now the United States, gogs means "a dog", whereas gogogs is used to refer to several dogs. This phenomenon, in which an element of a root word is repeated, sometimes with modification, within another word that is related to the root word, is called reduplication. In this case,the element "go" in gogs gets repeated in gogogs. There are many examples of this type of reduplication in O'odham.

The text makes which point about the O'odham word *gogogs*?

A) It contains a repetition of the element "go" in gogs.
B) It doesn't have a clear equivalent in English.
C) It is the only word in O'odham that uses reduplication.
D) It is identical in meaning to several other words in O'odham.

10

Master and Commander, first published in 1969, is a novel in Patrick O'Brian's Aubrey Maturin series, which includes twenty completed books. Some critics have found fault with the abrupt endings of Master and Commander and other books in the series, saying that they do not finish conclusively but arbitrarily stop. Other critics, however, argue that the books should not be thought of as discrete texts with traditional beginnings and endings but as a single incredibly long work, similar to other multivolume stories, such as John Galsworthy's The Forsyte Saga.

Which choice best states the main purpose of the text?

A) The unusual structure that O'Brian uses for Master and Commander makes it one of his most intricate books.
B) Critics have differing views regarding the efficacy of the structures of the novels in the Aubrey Maturin series.
C) Some critics think the Aubrey/Maturin series should have the literary renown of The Forsyte Saga, while others disagree.
D) Many critics judge the Aubrey/Maturin novels to be remarkably entertaining despite flaws in the novels' structures.

11

"The monster" is an 1898 story by Stephen Crane. In the story, the character of Jim, a young boy, accidentally damages a peony (a flower) in the yard while his father is tending to the lawn. Crane depicts the dedication and care with which his father typically cares for his lawn, writing, ______

Which quotation from "The Monster" most effectively illustrates the claim?

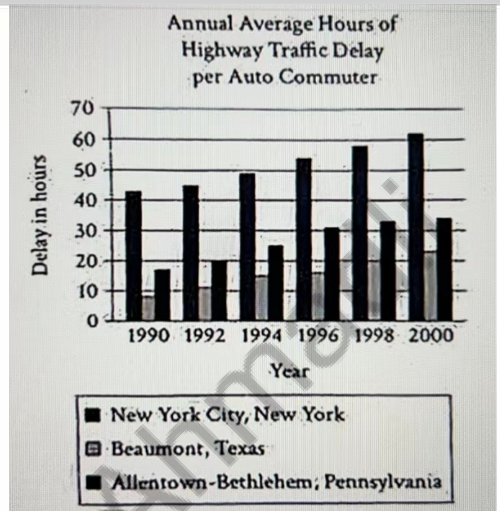
A) "After some trouble [Jim's father] found the subject of the incident, the broken flower. Turning then,he saw the child lurking at the rear and scanning his countenance."

B) "[Jim's father] was shaving this lawn as if it were a priest's chin. All during the season he had worked at it in the coolness and peace of the evenings after supper. Even in the shadow of the cherry-trees the grass was strong and healthy."

C) "[Jim's father] paused, and with the howl of the machine no longer occupying the sense, one could hear the robins in the cherry trees arranging their affairs."

D) "[Jim] went on to the lawn, very slowly, and kicking wretchedly at the turf. Presently his father came along with the whirring machine, while the sweet, new grass blades spun from the knives."

12



In a college course on urban affairs, a student asserts that increased traffic congestion in the 1990s in the United States was present both in very large cities such as New York City, New York, and smaller areas such as Allentown-Bethlehem, Pennsylvania, and Beaumont, Texas; though those smaller areas may have been less affected by traffic congestion than very large cities, this congestion also worsened in them over time.

Which choice best describes data from the graph that support the student's claim?

A) While the number of hours of traffic delay per commuter per year was always lower in the Allentown-Bethlehem, Pennsylvania area than in the New York City, New York area for each year between 1990 and 2000, the amount of traffic delay rose in both areas during this period.

B) In at least one of the three urban areas shown, the amount of traffic delay was less than 20 hours per person per year at one point between 1990 and 2000.

C) In 1992, the amount of traffic delay in the New York City, New York area was less than 20 hours per commuter per year.

D) Throughout the period between 1990 and 200, the annual amount of traffic delay per commuter was greater in the Allentown-Bethlehem, Pennsylvania area than in the New York City, New York area.

CONTINUE

13

Biologist Rosanna Alegado believes that we might learn how multicellular organisms developed from single-celled ones if we understand why the single-celled organism Salpingoeca rosetta, the oldest living relative of animals, sometimes forms colonies of cells. Alegado and colleagues reviewed data from many studies of how S, rosetta responds when exposed to another type of single-celled organism, bacteria, including John P. Bowman's work with Algoriphagus ratkowskyi bacteria and Iftikhar Ahmed's work with Algoriphagus boritolerans bacteria. Alegado and colleagues concluded that both A, ratkowskyi and A, boritolerans might have played a role in the development of multicellular organisms.

Which finding, if true, would most directly support Alegado and colleagues' conclusion?

A) Bowman and Ahmed found that S. rosetta tended to form colonies after bacterial exposure.
B) Bowman found that S. rosetta tended to form colonies after bacterial exposure,but Ahmed did not.
C) Ahmed found that S rosetta tended to form colonies after bacterial exposure, but Bowman did not.
D) Neither Bowman nor Ahmed found that s rosetta tended to form colonies after bacterial exposure.

14

Missing the question

15

Previous research has shown that plant species with a narrow geographical range tend to be more genetically homogeneous than plant species with extensive ranges are. Based on these findings, researchers recently ran simulations to predict how the genetic variation of several species of Mammillaria , a genus of cactus found throughout the Americas, might change in different distribution conditions. One of these species, M. anniana, is found only in the state of Tamaulipas. The researchers simulated what would happen if M. anniana spread to new habitats outside Tamaulipas, and, consistent with previous findings, the results showed that____

Which choice most logically completes the text?

A) there was a gradual increase in the genetic homogeneity of Mammillaria species in states neighboring Tamaulipas.

B) Mammillaria species other than M. anniana would become more common in Tamaulipas.

C) several other Mammillaria species could survive in Tamaulipas in the future.

D) the genetic homogeneity of M. anniana decreased over time.

16

In the periodic table, an element's atomic number indicates the number of protons in an atom of that element. For example, there are 25 protons in a manganese atom. Raymond Chang's textbook General Chemistry: The Essential Concepts ____ this concept in detail.

Which choice completes the text so that it conforms to the conventions of standard English?

A) discussing
B) discusses
C) to discuss
D) having discussed

17

An emulsifier is a type of compound that ___ to stabilize an emulsion--a mixture of two or more liquids that otherwise would not easily blend together. In the cosmetics industry, emulsifiers like cetostearyl alcohol are commonly used to blend oil and water into homogeneous formulations, like lotions and perfumes.

Which choice completes the text so that it conforms to the conventions of standard English?

A）served
B）had served
C）serves
D）was serving

18

Over the years, dozens of architectural and archaeological sites important to Hawaiian history and culture--such as Grove Farm, Kilauea School, and the Hana Belt Road--have been added to an important ___ Hawai'i Register of Historic Places. New sites are added each year, as decided by a review board of experts including sociologist Alton Okinaka, architect Katle Stephens, and historian Bill Souza.

Which choice completes the text so that it conforms to the conventions of standard English?

A) list: the
B) list. The
C) list; the
D) list the

19

Today, the Michelin Guide is widely known as the arbiter of fine dining, but when it was created in 1990, it was little more than a marketing gimmick. Brothers Andre and Edouard Michelin sought to increase profits for their tire company by encouraging their customers to drive across France, visiting the guide's recommended hotels and ____as it grew in scope and influence, the guide had it's modest French eateries replaced with the best restaurants from around the world, including The Eight in Macau and Sushi Ueda in Nagoya.

Which choice completes the text so that it conforms to the conventions of standard English?

A) restaurants
B) restaurants;
C) restaurants and
D) restaurants,

20

In their research, behavioral ___ integrate methods from psychology and economics to analyze how and why people make particular choices and to examine the broader implications of those choices for the economy.

Which choice completes the text so that it conforms to the conventions of standard English?

A) economists Laura Gee and Muriel Niederle,
B) economists Laura Gee and Muriel Niederle
C) economists Laura Gee, and Muriel Niederle,
D) economists, Laura Gee and Muriel Niederle,

21

The city of Jinan, China, expanded its rapid transit system, the Jinan Metro, as recently as 2020. ____the system will likely need to undergo further expansions as the size and needs of the city's population shift.

Which choice completes the text with the most logical transition?

A) Still,
B) By contrast,
C) For example,
D) Instead,

22

Theater industry veteran Robert Tang admits that his ambitions weren't very lofty when he created the Asian American Theater Revue, a website that documents information about Asian American theaters like Second Generation Productions in New York City, New York: "Everyone had a site, so I had to have one, too." ____ his site has become an essential resource for industry professionals looking to learn more about Asian American plays, playwrights, theater festivals, and more.

Which choice completes the text with the most logical transition?

A) Nevertheless,
B) Therefore,
C) Furthermore,
D) Next,

23

The first documented use of the English word "session" is attributed to poet Geoffrey Chaucer's 1386 work "Canterbury Prologue." However, Chaucer didn't write in Modern English; ___he wrote in what we now call Middle English, which was commonly used during the period.

Which choice completes the text with the most logical transition?

A) rather,
B) as a result,
C) similarly,
D) finally,

24

While researching a topic, a student has taken the following notes:

- Kale is a vegetable that contains ascorbic acid, an essential nutrient for humans.
- Blueberries are fruits that contain ascorbic acid.
- There is 120 milligrams (mg) of ascorbic acid per every 100 grams (g) of kale.
- There is 10 mg of ascorbic acid per every 100g of blueberries.
- Humans cannot make ascorbic acid in their bodies, so they must get it from foods, including fruits and vegetables.
- Ascorbic acid is also known as vitamin C.

The student wants to make and support a claim that kale is a better source of vitamin C than blueberries are. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) Kale contains ascorbic acid (which humans cannot make in their bodies), and blueberries do too.
B) Blueberries contain 10 mg of vitamin C per 100 g, but kale is a better source of vitamin C; in fact, kale contains 120 mg per 100 g.
C) Kale is a better source of vitamin C, also known as ascorbic acid, than blueberries are.
D) Humans must get vitamin C from foods like kale and blueberries because they cannot make it in their bodies.

25

While researching a topic, a student has taken the following notes:

- The Mohs scale of mineral hardness is a ten-point scale that orders minerals by hardness based on their ability to scratch other minerals.
- Minerals with larger numbers are harder than minerals with smaller numbers and can leave visible scratches on them.
- Minerals with smaller numbers are softer than minerals with lager numbers and cannot leave visible scratches on them.
- The mineral calcite has a Mohs scale number of 3.
- The mineral apatite has a Mohs scale number of 5.
- The mineral topaz has a Mohs scale number of 8.

The student wants to explain how hard apatite is in relation to other minerals. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) In the Mohs scale of mineral hardness, topaz (8) is ranked higher than calcite (3).

B) Calcite, topaz, and apatite can be ordered by their ability to leave visible scratches on other minerals.

C) Topaz has a Mohs scale number of 8, which means that it can scratch not only calcite but also apatite.

D) Apatite has a Mohs scale number of 5, which means that it is harder than calcite (3) but softer than topaz (8).

26

While researching a topic, a student has taken the following notes:

- Epistolary novels are novels written primarily as a series of fictional documents, typically letters.
- Sometimes, the documents are journal entries, newspaper clippings, and more.
- Julie, or The New Heloise (1761) is an epistolary novel by French author Jean-Jacques Rousseau.
- It consists primarily of letters sent between a man and a woman living in a small French town.
- Absolutely Normal Chaos (1990) is an epistolary novel by American author Sharon Creech.
- It consists primarily of journal entries written by a student during her summer vacation.

The student wants to emphasize a difference between the two novels. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) The French author Jean-Jacques Rousseau and the American author Sharon Creech have written epistolary novels.

B) While Julie, or The New Heloise and Absolutely Normal Chaos are both epistolary novels, the former is composed primarily of letters and the latter of journal entries.

C) Some epistolary novels consist primarily or journal entries, but most, like Jean-Jacques Rousseau's Julie, or The New Heloise, consist primarily of letters.

D) Jean-Jacques Rousseau's Julie, or The New Heloise and Sharon Creech's Absolutely Normal Chaos demonstrate the variety of documents that can be used in epistolary novels.

27

While researching a topic, a student has taken the following notes:

- Most of the plant and bird species in Oahu, Hawaii, are non-native.
- In a 2019 study, researchers wanted to know what role non-native birds play in dispersing plant seeds in Oahu.
- Researchers catalogued plant seeds found in fecal samples from nonnative birds.
- Leptecophylla tameiameiae, a flowering shrub, was one of fifteen native species catalogued.
- Passiflora suberosa, an herbaceous vine, was one of twenty-nine nonnative species catalogued.
- Researchers concluded that non-native birds play a vital role in dispersing the seeds of native and non-native plants.

The student wants to compare the number of native and non-native species catalogued. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) Seeds form Leptecophylla tameiameiae and Passiflora suberosa can be found in Oahu, Hawaii, but only the former plant in native.

B) In 2019, researchers catalogued native and non-native plant seeds found in fecal samples from non-native birds.

C) Most of the birds in Oahu, Hawaii, are non-native, and researchers have concluded that they play a vital role in seed dispersal.

D) When cataloguing plant seeds found in bird fecal samples, researchers found more non-native seed species than native species.

No Test Material On This Page

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33 QUESTIONS

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1

Whether the reign of a French monarch such as Louis VII or Henry V was historically consequential or relatively uneventful, its trajectory was shaped by questions of legitimacy and therefore cannot be understood without a corollary understanding of the characteristics without which the monarchwould have been forced to ___ the throne.

Which choice completes the text with the most logical and precise word or phrase?

A) reciprocate
B) annotate
C) abandon
D) equalize

2

With his widely celebrated portraits of poet Seamus Heaney, molecular biologist Struther Arnott, and other prominent figures in different fields, British painter Peter Edwards has ___ substantial prestige as an artist.

Which choice completes the text with the most logical and precise word or phrase?

A) required
B) remembered
C) achieved
D) avoided

3

Though many ___ studies of the effect of altitude on blood chemistry focus on people who live above sea level, researchers Suleiman A. Al-Sweedan and Moath Alhaj recently chose the novel path of focusing on people who live below sea level, in locations such as the California towns of Salton Cityand Coachella.

Which choice completes the text with the most logical and precise word or phrase?

A) eccentric
B) meager
C) conventional
D) random

4

The following text is adapted from William Wordsworth's 1798 poem "Lines Written a Few Miles above Tintern Abbey."

Once again
Do I behold these steep and lofty cliffs,
Which on a wild secluded scene impress
Thoughts of more deep seclusion; and connect
The landscape with the quiet of the sky.

As used in the text, what does the word "deep" most nearly mean?

A) Tough
B) Hideous
C) Complicated
D) Intense

5

The Caves of Gargas paintings--stencils of human hands found in what is now France and dating from around 27000 years ago--are thought of as art today, but the question of whether the people of the time understood the paintings as something akin to art in our modern sense or in some other way entirely is ___; we will never be able to answer it.

Which choice completes the text with the most logical and precise word or phrase?

A) irresolvable
B) self-contradictory
C) imperative
D) unavoidable

6

Eden Robinson is a novelist and a member of the Haisla Nation in western Canada. Critics and fellow writers have praised her work for combining traditional Haisla stories with popular genres of literature, such as fantasy and mystery fiction. But Robinson is not the only Indigenous writer to blend traditional stories with popular literature. In the 2019 novel Empire of wild, Cherie Dimaline successfully blended the oral storytelling tradition of her people, the Metis, with horror fiction.

Which choice best states the main purpose of the text?

A) To discuss two writers who blend traditional Indigenous stories with popular genres
B) To urge younger Indigenous authors to avoid writing in popular genres
C) To argue that traditional Indigenous stories are more memorable than most novels are
D) To explain why one Indigenous author has achieved more success than another has

7

Text 1

<u>Attempts to automate classification of music into genres have not been very successful, and we may be at the limit of what is technologically possible.</u> But it's not clear that this is a worthwhile pursuit in any case--as Jin Ha Lee and Anh Thu Nguyen argue in their study of the South Korean band BTS, relationships between pieces of music may be best understood with concepts other than genre.

Text 2

Tango is a genre of music originally from Argentina and Uruguay that shares some harmonic and rhythmic similarities with the pagode genre. Automated genre classification systems typically struggle to draw distinctions in situations like this, but Yandre Costa and colleagues solved that problem by converting sound to images and having computers compare features of those images, an approach that demonstrates how much innovation is possible in this field.

Based on the texts, how would the author of Text 2 most likely respond to the claim in the underline sentence of Text 1?

A) By arguing that people tend to disagree when identifying genre classifications for music

B) By suggesting that the concept of genre may become more useful for music listeners

C) By criticizing previous research into automated music genre classification for favoring specific genre categories

D) By asserting that it may be possible to improve automated classification systems

8

The following text is adapted from Adib Khorram's 2018 novel Darius the Great Is Not Okay. Darius, a teenager from the United States, is visiting his family members in Iran.

> I dozed and floated on the clouds of Farsi that blew my way from the front seat of Dayi [Uncle] Jamsheed's SUV.
> It reminded me of when I was little, and Mom chanted to me in Farsi every night before bedtime. It's hard to describe Farsi chanting: the way Mom drew her voice out like the notes of a cello as she recited poems by Rumi or Hafez I didn't know what they meant, but that didn't matter. It was quiet and soothing.

©2018 by Adib Khorram

Base on the text, what does the narrator mainly remember about the times when his mother chanted to him?

A) That he liked the characters in the stories his mother made up while chanting

B) That his mother had music playing in the background while she chanted

C) That his mother didn't usually chant at night

D) That he found the sound of his mother's chanting to be calming

9

Horses and barn owls can see in three dimensions (3D), which helps them perceive distance and depth. Octopuses and squid are thought to lack 3D vision. many researchers once thought the same about cuttlefish, but Trevor Wardill and his team wanted to test whether this assumption was true. The team studied how cuttlefish wearing 3D glasses reacted to 3D images of shrimp (a favorite prey) projected on a tank wall. Cuttlefish changed their striking position to match the 3D images, suggesting that their vision is more like that of horses and barn owls than that of octopuses or squid.

Which choice best states the main idea of the text?

A) Researchers have long known that horses and barn owls can see in 3D.
B) Cuttlefish are surprisingly similar in structure to octopuses and squid.
C) Contrary to what many researchers had assumed, cuttlefish may be able to see in 3D.
D) The ability to see in 3D allows many animals to interact with one another.

10

Minimum and Maximum Depths of Stony Coral Species in Caribbean and Indo-Pacific Waters

Species	Minimum depth (meters)	Maximum depth (meters)	Range (meters)
Acropora anthocercis	5	10	5
Cyphastrea hexasepta	8	25	17
Agaricia fragilis	10	102	92
Heliofungia fralinae	3	27	24

The table is from a 2018 study in which Luiz Rocha and colleagues examined the ranges of depths at which certain stony coral species have been found in Caribbean and Indo-Pacific waters. Among the corals in the table. the species with the greatest range between minimum and maximum depths is___

Which choice most effectively uses data from the table to complete the statement?

A) Acropora anthocercis.
B) Cyphastrea hexasepta.
C) Agaricia fragilis.
D) Heliofungia fralinae.

11

The Underdogs is a 1915 novel by Mariano Azuela, originally written in Spanish. In the novel, a group of soldiers travel through a canyon, where their collective mood becomes strongly affected by the strenuous conditions of their journey: _____

Which quotation from a translation of *The Underdogs* most effectively illustrates the claim?

A) "The sierra is clad in gala colors. Over its inaccessible peaks the opalescent fog settles like a snowy veil on the forehead of a bride."
B) "All day long [the soldiers] rode through the canyon, up and down the steep, round hills, dirty and bald as a man's head, hill after hill in endless succession."
C) "Then, hurriedly, [the soldiers] took the Juchipila canyon northward, without halting to rest until nightfall."
D) "The sun, beating down upon [the soldiers], dulled their minds and bodies and presently they were silent."

12

US Hydroelectric Power Plants, 2019

Plant	State	Mode	Average power generation(MWh/yr)	Water source
Scanlon	Minnesota	run-of-river	7511	St. Louis River
Kansas River	Kansas	run-of-river	15345	Kansas River
Squa Pan Hydro Station	Maine	peaking	881	Squa Pan Stream
Great Falls	Tennessee	peaking	124392	Caney Fork River

A run-of-river hydroelectric power plant, as the name suggests, uses the natural flow of a water source to generate electricity but is unable to start or stop that flow through its generators. In contrast, a peaking hydroelectric power plant (used when demand for electricity peaks) controls the flow of water through its generators: starting flow when demand is high enough, stopping it when demand is too low, and otherwise regulating it to keep pace with changing electricity needs. Although peaking plants do not typically operate continuously as run-of-river plants do, peaking plants can generate more megawatt-hours of power per year (MWh/yr) than some run-of-river plants. For example, the _____

Which choice most effectively uses data from the table to complete the example?

A) average power generated annually by the Great Falls plant is higher than that generated by any of the run-of-river plants in the table.
B) Scanlon plant, which is a run-of-river plant, has more generators than any of the other plants in the table.
C) run-of-river plant with the highest average annual power generation in the table generates more electricity than the peaking plant with the highest annual power generation in the table.
D) average power generated annually by the Kansas River plant is higher than that generated by the Scanlon plant.

13

The 2000 production of The Green Bird was the first Broadway show for which Constance Hoffman was credited as a costume designer. Hoffman was among the Broadway costume designers interviewed by Sara Jablon-Roberts and Eulanda A Sanders for their study of historical accuracy in costume design for shows with a historical setting. They found that even designers who value historical accuracy will often include contemporary design elements that don't fit with the historical period. In a research paper about theatrical costume design, a student argues that costume designers for modern productions of Shakespeare's A Midsummer Night's Dream (set in ancient Greece) sometimes include such elements unintentionally.

Which quotation from an interview with a costume designer would most effectively support the student's claim?

A) "I tend to pay careful attention to a character's accessories like gloves, hats, and jackets. These elements help communicate information about how this person would have fit into society at the time."

B) "I aim to create a clear sense of the character and the world they inhabit. Sometimes this means adhering to the historical period's style, but frequently, as in stagings of A Midsummer Night's Dream, some flexibility is required to communicate an idea to the audience."

C) "Costumes must always reflect the correct historical period. Too often, I've seen costumes that borrow from a wide span of periods, resulting in a production with a confusing visual style."

D) "In the 1980s I was costuming for a production of A Midsummer Night's Dream. Even though I focused on using the correct materials and techniques for the play's period in history, when I look back I can clearly see the influence of 1980s fashion in my designs."

14

In order to identify research trends, Shirley Ann Williams et al, reviewed a collection of studies of the social media website Twitter, such as the 2011 paper by Celik et al, titled "Learning Semantic Relationships between Entities in Twitter." Williams's team searched for the term "Twitter" on the Scopus and Web of Science databases and found that, though most papers returned by the search did in fact focus on the social media platform, a few discussed unrelated subjects such as sounds made by tractor engines. One reasonable explanation for this result is that it occurred because_____

Which choice most logically completes the text?

A) a group of researchers had an extensive discussion on Twitter about the design of tractor engines.

B) most of the papers in the Scopus and Web of Science databases do not discuss automotive engines or social media websites.

C) the word "twitter" may not always refer to a social media website, but to other subjects such as noises made by machines.

D) academics who are active on social networks are likely to announce their recent publications on Twitter.

15

In a 2012 study in the United States, Michael E. Berndt and Travis K. Bavin found a positive association between levels of dissolved organic carbon and mercury in bodies of fresh water. Many other studies have yielded similar results, suggesting to some scientists that this association is true for all bodies of fresh water. But much of that research has been conducted at broadly similar sites in North America, and when Seam Noh and colleagues examined bodies of fresh water in South Korea, they found a negative association between dissolved organic carbon levels and mercury levels. If similar findings emerge from other locations outside North America, that could suggest that _____

Which choice most logically completes the text?

A) the mercury levels reported in Noh and colleagues' study were much higher than those reported in the study by Brebdt and Bavin even though the dissolved organic carbon levels reported in the two studies were approximately the same.
B) Berndt and Bavin may have inadvertently measured a different characteristic of bodies of fresh water than their levels of dissolved organic carbon and mercury.
C) the relationship between dissolved organic carbon and mercury reported by Berndt and Bavin reflects conditions that are characteristic of certain kinds of ecosystems in North America rather than universal conditions.
D) most of the studies conducted in North America have measured dissolved organic carbon and mercury levels at a higher level of precision than was the case in Noh and colleagues' study.

16

In the thought-provoking 2010 exhibition Relative Pelican at the Stux Gallery in New York, Sokari Douglas Camp presented sculptures that had been ___ welded from recycled metal materials, such as oil drums. In doing so, the London-based Nigerian sculptor challenged viewers to imagine new uses and futures for the material wastes of the past.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) cut, bent, and,
B) cut, bent, and
C) cut, bent; and
D) cut bent, and

17

In Puerto Rico, it's not unusual for a city or town to be known____ a nickname that corresponds to one of its notable features, like landscape, climate, famous residents, or chief exports. For example, the Puerto Rican municipality of Hatillo has also been called "the Land of Green Fields," a nickname that alludes to what the area is well known for: the lush greenery that surrounds it.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) by
B) by-
C) by,
D) by:

18

In 2021, Sarah McAnulty ___as a biologist at the University of Connecticut, studying squids.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) working
B) was working
C) to work
D) to have worked

19

The word "robot" ____ first used in the 1920 play R.U.R. by Czech author Karel Capek.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) were
B) were being
C) was
D) have been

20

In the spring of 1964, it was hard to go anywhere without hearing the song "Cotton Candy" by the Sylvers. As of April 11, the song had spent thirteen straight weeks on the Billboard Hot 100 chart and ___ No.1 on the list.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) have been ranking
B) was ranked
C) were ranked
D) are ranking

21

In 2000, the United States Congress designated a portion of Pennsylvania as the Lackawanna Valley National Heritage Area, _____ the region's unique natural, cultural, and historical attributes.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) they cite
B) cited
C) citing
D) has cited

22

In 1940, scientists first isolated a sample of the steroid hormone androstenone, which they knew existed in animals--and, they assumed, only in animals. ____ scientists know that androstenone can also be found in chokecherry plants (Prunus virginiana), but this discovery didn't occur until decades later.

Which choice completes the text with the most logical transition?

A) Today,
B) For instance,
C) In other words,
D) Thus,

23

According to traditional classifications, each of William Shakespeare's plays belongs to one of three distinct genres: comedy, history, or tragedy. The play Much Ado About Nothing (1598-99), ____ is classified as a comedy, whereas Richard II (1595) is classified as a history.

Which choice completes the text with the most logical transition?

A) in addition,
B) similarly,
C) for instance,
D) however,

24

If you were to view the Ol Kokwe volcano in Kenya or the Asavyo volcano in Ethiopia from above, you might notice that their low profiles and gently sloping sides make each of them look a bit like a shield lying flat on the ground. _____ volcanologists classify volcanoes of this type as shield volcanoes.

Which choice completes the text with the most logical transition?

A) Still,
B) Besides,
C) Second,
D) Fittingly,

25

Though its onboard laboratory allowed it to analyze rock samples on site, the 2011 Mars Curiosity rover was unable to preserve samples for future analysis. ____ when creating the 2020 Mars Perseverance rover, robotics mechanical engineer Arash Kalantari and other members of NASA's Jet Propulsion Laboratory sought to implement a mechanism that could do exactly that.

Which choice completes the text with the most logical transition?

A) Previously,
B) Specifically,
C) Likewise,
D) Thus,

26

Though the musical groups Los Palominos and Banda Los Recoditos are often categorized under the catchall "regional Mexican music" label, they each adhere to distinct musical traditions. The former works within the tejano tradition, which originated in mid-twentieth-century Texas and northern Mexico. The latter,____ follows the banda tradition, with roots in min-nineteenth-century southern and central Mexico.

Which choice completes the text with the most logical transition?

A) therefore,
B) to this end,
C) however,
D) in other words,

27

Water from the aquifers of Alaska's Susitna Glacier travels the 160 km length of the Skwentna River before flowing into the Yentna River and then reaching the mouth of the river system's main stem, the Susitna River. _____ at the Susitna's mouth, hundreds of kilometers from the source, the water empties into the Gulf of Alaska.

Which choice completes the text with the most logical transition?

A) There,
B) For instance,
C) Regardless,
D) Instead,

No Test Material On This Page

Math

27 QUESTIONS

DIRECTIONS

The questions in this section address a number of important math skills.
Use of a calculator is permitted for all questions.

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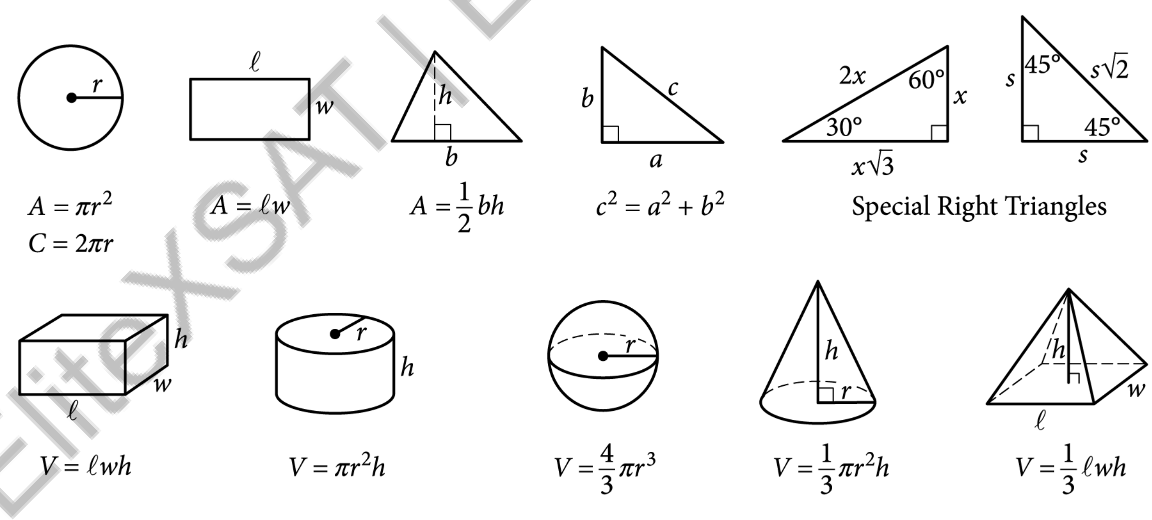

The number of degrees of arc in a circle is 360.

The number of radians of arc in a circle is 2π.

The sum of the measures in degrees of the angles of a triangle is 180.

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- Don't include **symbols** such as a percent sign, comma, or dollar sign in your circled answer.

1

The table summarizes the UV index value recorded by a research assistant at noon each day for 49 days.

UV index	Number of days
1	9
2	15
3	12
4	13

According to the table, a UV index value of 1 was recorded on how many days?

A) 4

B) 9

C) 40

D) 49

2

To make a bookcase, a woodworker charged a onetime fee plus \$17 per hour worked. The equation $17h+45=164$ represents this situation, where h is the number of hours worked. Which of the following is the best interpretation of 164 in this context?

A) The onetime fee, in dollars

B) The number of hours worked

C) The charge per hour, in dollars

D) The total charge, in dollars

3

ln December 2017, the lowest temperature recorded in a certain city was 40 degrees Fahrenheit (°F) and the highest temperature recorded was 90 °F. Which inequality is true for all values of t, where t represents any temperature, in °F, recorded in the city in December 2017?

A) $40\leq t\leq 90$

B) $t\leq 40$

C) $t\leq 50$

D) $t\geq 90$

4

The function f is defined by $f(x)=6(2x+4)$. For what value of x does $f(x)=48$?

A) 2

B) 6

C) 8

D) 22

5

- If your answer is a fraction that doesn't fit in the provided space, enter the decimal equivalent.
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- Don't enter symbols such as a percent sign, comma, or dollar sign.

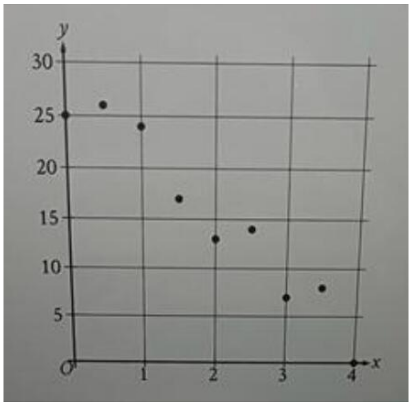
Answer	Acceptable ways to enter answer	Unacceptable: will NOT receive credit
3.5	3.5 3.50 $\frac{2}{7}$	31/2 3 1/2
$\frac{2}{3}$	2/3 .6666 .6667 0.666 0.667	0.66 .66 0.67 .67
$-\frac{1}{3}$	-1/3 -.3333 -0.333	-.33 -0.33

The ratio of green tiles to blue tiles in a piece of artwork is 5 to 2. If there are 16 blue tiles in the piece of artwork, how many green tiles are there?

Answer : ___

6

The scatterplot shows the relationship between two variables, x and y.



Which of the following equations is the most appropriate linear model for the data shown?

A) $y = -6 + 28x$

B) $y = 6 - 28x$

C) $y = 28 + 6x$

D) $y = 28 - 6x$

7

A scientist analyzed a soil sample with a mass of 900 grams and determined that it contained 189 grams of water. What is the percentage of water, by mass, in this soil sample?

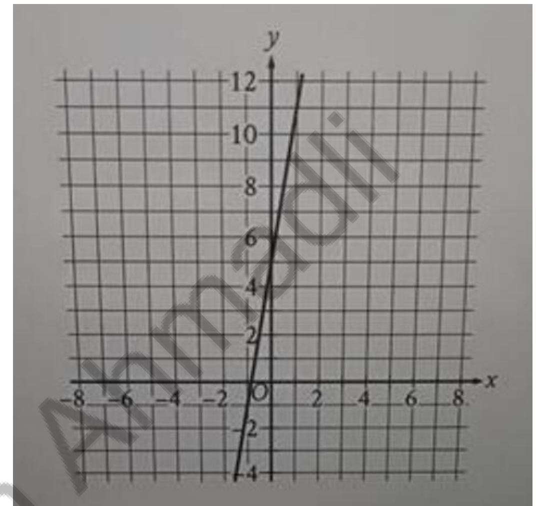
A) 9%

B) 9.9%

C) 18.9%

D) 21%

8



Line n is shown in the xy-plane. Line k (not shown) is perpendicular to line n. What is the slope of line k?

A) -1/5

B) -1/6

C) 5

D) 6

9

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$$mx + ky = -83$$
$$2x + ky = 22$$

In the given system of equations, m and k are constants. The graphs of these equations in the xy-plane intersect at the point $(5, y)$. What is the value of m?

Answer: -19

10

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$\frac{2}{3}$	2/3 .6666 .6667 0.666 0.667	0.66 .66 0.67 .67
$-\frac{1}{3}$	-1/3 -.3333 -0.333	-.33 -0.33

For the linear function f, the graph of $y = f(x)$ in the xy-plane passes through the points $(0, 2)$ and $(3, 3)$. what is the slope of $y = f(x)$.

Answer: 1/3

CONTINUE

11

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A certain book has 250 pages, and 21 of these pages have an illustration. If one of the book's pages is selected at random, what is the probability of selecting a page with an illustration? (Express your answer as a decimal of fraction, not as a percent.)

Answer : 21/250

12

$$b^2 + 5c = 9d$$

The given equation relates the real numbers b, c, and d, where $d > 5/9c$. Which equation correctly expresses b in terms of c and d?

A) $b = \frac{9d + 5c}{2}$

B) $b = \frac{9d - 5c}{2}$

C) $b = \pm\sqrt{9d + 5c}$

D) $b = \pm\sqrt{9d - 5c}$

13

A right circular cylinder has a height of 4 meters (m) and a base with a radius of 18 m. What is the volume, in m^3, of the cylinder?

A) 4π
B) 22π
C) 288π
D) 1296π

14

What is the y-intercept of the graph of $3x + 2y = 96$ in the xy-plane?

A) (0, 5)
B) (0, 6)
C) (0, 32)
D) (0, 48)

15

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If $\frac{6}{7}p + 12 = 54$, what is the value of $7p$?

Answer : 343

16

A certain neighborhood had a population of 1340 in 2009. Each year for the next 5 years, the population of the neighborhood increased by approximately 3% of the population the previous year. Which of the following equations represents the population, N, of the neighborhood t years after 2009, where $t \leqslant 5$?

A) $N = 0.03(1340)^t$

B) $N = 1340(0.03)^t$

C) $N = 1340(1.03)^t$

D) $N = 1.03(1340)^t$

17

In the xy-plane, which of the following does NOT contain any points (x, y) that are solutions to $7x + 4y > 12$?

A) The region where $x > 0$ and $y > 0$

B) The region where $x < 0$ and $y > 0$

C) The region where $x < 0$ and $y < 0$

D) The region where $x > 0$ and $y < 0$

18

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In triangle *RST*, the measure of angle *R* is 10 degrees and the measure of angle *T* is 50 degrees. Point *L* lies on *RS*, point K lies on *ST*, and *LK* is parallel to *RT*. What is the measure, in degrees, of angle *SKL*? (Disregard the degree symbol when entering your answer.)

Answer : 50

19

An auditorium has seats for 3200 people. Tickets to attend a show at the auditorium currently cost \$8.00. For each \$1.00 increase to the ticket price, 100 fewer tickets will be sold. This situation can be modeled by the equation $y = -100x^2 + 2400x + 25600$, where x represents the increase in ticket price, in dollars, and y represents the revenue, in dollars, from ticket sales. If this equation is graphed in the xy-plane, at what value of x is the maximum of the graph?

A) 8
B) 12
C) 24
D) 32

20

$$(x + 3)^2 + (y - 4)^2 = 25$$

In the xy-plane, the graph of the given equation is a circle. Which point lies on this circle?

A) $(-3, 4)$
B) $(3, -4)$
C) $(\sqrt{11} + 3, \sqrt{14} - 4)$
D) $(\sqrt{11} - 3, \sqrt{14} + 4)$

21

The expression $\dfrac{x^{20}(x - 4)}{5x^2} + \dfrac{4x^{20}}{5x^2}$ is equivalent to $1/5x^c$, where c is a constant and $x > 0$. What is the value of c?

A) 4
B) 5
C) 19
D) 21

22

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Triangle ABC is similar to triangle DEF, where angle A corresponds to angle D and angles C and F are right angle. The length of AB is 2.4 times the length of DE. If tan A =21/20, what is the value of sin D?

Answer :

No Test Material On This Page

Math

27 QUESTIONS

DIRECTIONS

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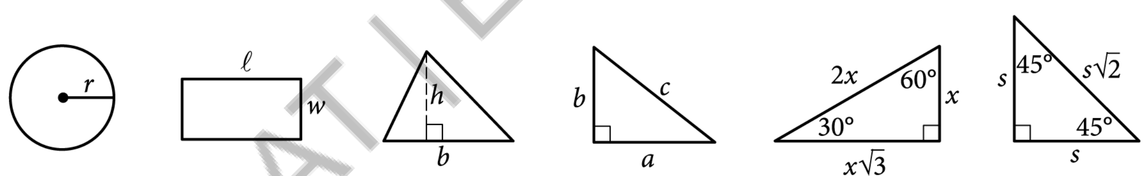


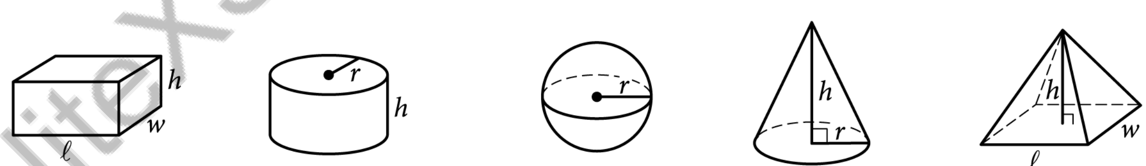
$A = \pi r^2$
$C = 2\pi r$

$A = \ell w$

$A = \frac{1}{2}bh$

$c^2 = a^2 + b^2$

Special Right Triangles



$V = \ell wh$

$V = \pi r^2 h$

$V = \frac{4}{3}\pi r^3$

$V = \frac{1}{3}\pi r^2 h$

$V = \frac{1}{3}\ell wh$

The number of degrees of arc in a circle is 360.

The number of radians of arc in a circle is 2π.

The sum of the measures in degrees of the angles of a triangle is 180.

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1

The speed of a white-throated needletail, a type of bird, in flight was measured to be 49 miles per hour. What was the white-throated needletail's measured speed, in kilometers per hour? (Use 1 mile = 1.6 kilometers.)

A) 30.6

B) 47.4

C) 50.6

D) 78.4

2

Which expression is equivalent to $(3x^3 - x^2 + 4)(5x^2 + 8x)$?

A) $15x^5 + 19x^4 - 8x^3 - 20x + 32$

B) $15x^5 + 19x^4 - 8x^3 - 20x^2 + 32$

C) $15x^5 + 19x^4 - 8x^3 + 20x^2 + 32x$

D) $15x^5 + 29x^4 - 8x^3 + 20x^2 + 32x$

3

If $5(x + 1) = 25$, what is the value of $x + 1$?

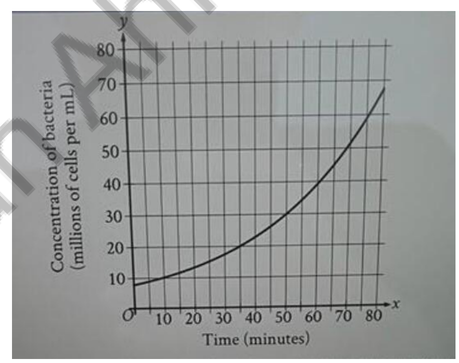
A) 4

B) 5

C) 19

D) 20

4



The graph shows the estimated concentration of a certain strain of bacteria, in millions of cells per mL of nutrient medium, over time x, in minutes since the initial measurement. According to the graph, which of the following is closest to the number of minutes in took for the estimated concentration of the bacteria to increase from 20 million cells per mL of nutrient medium to 30 million cells per mL of nutrient medium?

A) 15

B) 20

C) 30

D) 35

5

If $(x+3)^2 = 30$, what is the value of $x^2 + 6x$?

A) 21

B) 30

C) 39

D) 60

6

$$y = 5x$$
$$y = 2x + 2$$

How many solutions does the given system of equations have?

A) Exactly one

B) Exactly two

C) Infinitely many

D) Zero

7

$$f(x) = |71 - 2x|$$

The function f is defined by the given equation. For which of the following values of k does $f(k) = 3k$?

A) 71/5

B) 71/2

C) 213/5

D) 71

8

$$g(x) = 2(16x - 17)$$

What is the y-coordinate of the y-intercept of the graph of $y = g(x) - 3$ in the xy-plane?

A) -37

B) -34

C) -20

D) -17

9

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x	y
0	n
4	n + 19
8	n + 38

There is a linear relationship between x and y. The table shows three values of x and their corresponding values of y in terms of a constant n. What is the slope of the line that represents this relationship in the xy-plane?

Answer:

10

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$\frac{2}{3}$	2/3 .6666 .6667 0.666 0.667	0.66 .66 0.67 .67
$-\frac{1}{3}$	-1/3 -.3333 -0.333	-.33 -0.33

$$18x^2 + 24x + c = 0$$

In the given equation, c is a constant. The equation has exactly one solution. What is the value of c?

Answer:

11

- If your answer is a fraction that doesn't fit in the provided space, enter the decimal equivalent.
- If your answer is a decimal that doesn't fit in the provided space, enter it by truncating or rounding at the fourth digit.
- If your answer is a mixed number (such as $3\frac{1}{2}$), enter it as an improper fraction (7/2) or its decimal equivalent(3.5).
- Don't enter symbols such as a percent sign,comma, or dollar sign.

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$$(x - 3) - 8(y + 9) = 129$$

$$(x - 3) + 8(y + 9) = 432$$

The solution to the given system of equations is (x, y). What is the value of $8(x - 3)$?

Answer:

12

The volume of a right rectangular prism with a square base is 2448 cubic centimeters. If the area of the square base is 144 square centimeters, what is the area, in square centimeters, of one of the four lateral faces of the prism?

A) 17
B) 204
C) 540
D) 816

13

$$y = 5(x - 2)^2$$
$$y = 10(x - 2)$$

A solution to the given system of equations is (x, y). What is one possible value of $x + y$?

A) -20
B) 4
C) 20
D) 24

CONTINUE

14

• If your answer is a fraction that doesn't fit in the provided space, enter the decimal equivalent.

• If your answer is a decimal that doesn't fit in the provided space, enter it by truncating or rounding at the fourth digit.

• If your answer is a mixed number (such as $3\frac{1}{2}$), enter it as an improper fraction (7/2) or its decimal equivalent(3.5).

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The graph of the quadratic function $y = f(x)$ in the xy-plane intersects the x-axis when $x = 39$ and when $x = p$, where p is a constant. The maximum value of $y = f(x)$ occurs at the point $(14, m)$, where m is a constant. What is the value of p?

Answer:

15

• If your answer is a fraction that doesn't fit in the provided space, enter the decimal equivalent.

• If your answer is a decimal that doesn't fit in the provided space, enter it by truncating or rounding at the fourth digit.

• If your answer is a mixed number (such as $3\frac{1}{2}$), enter it as an improper fraction (7/2) or its decimal equivalent(3.5).

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$$\frac{x+1}{5x^2} = \frac{k}{x}$$

In the given equation, k is a constant. The solution to the given equation is 1/174. What is the value of k?

Answer:

16

- If your answer is a fraction that doesn't fit in the provided space, enter the decimal equivalent.

- If your answer is a decimal that doesn't fit in the provided space, enter it by truncating or rounding at the fourth digit.

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The area of a triangle is equal to x^2 square centimeters. The length of the base of the triangle is $2x + 6$ centimeters, and the height of the triangle is $x - 2$ centimeters. What is the value of x?

Answer:

17

The function g is a quadratic function. In the *xy*-plane, the graph of $y = g(x)$ has a vertex at (-1, -4) and passes through the points (-2, -43) and (1, -160). What is the value of $g(0) - g(2)$?

A) -121

B) 0

C) 117

D) 312

18

Triangles *ABC* and *DEF* are congruent, where *A* corresponds to *D*, and *B* and *E* are right angles. The measure of angle *A* is 62°. What is the measure of angle *F*?

A) 28°

B) 62°

C) 90°

D) 118°

19

A business consultant charges \$408 for the first hour and \$204 for each additional hour of work. Whcih of the following functions gives the charge *C(h)*, in dollars, for *h* hours of work, where *h* is a positive integer?

A) $C(h) = 204h + 204$

B) $C(h) = 204h + 408$

C) $C(h) = 408h + 204$

D) $C(h) = 408h + 612$

20

An acceptable noise criterion rating for the background noise in a laundry room is 50. For a noise criterion rating of 50, the equation $y = 22(0.997)^{x-60} + 47$ gives the estimated sound pressure level, y, in decibels, as a funciton of the octave band center frequency, x, in hertz, where $x \geq 60$. Which of the following is the best interpretation of 47 in this context?

A) 47 is 22 less than the estimated sound pressure level, in decibels, at an octave band center frequency of 60 hertz.

B) 47 is 22 less than the estimated sound pressure level, in decibels, at an octave band center frequency of 0 hertz.

C) 47 is the estimated sound pressure level, in decibels, at an octave band center frequency of 60 hertz.

D) 47 is the estimated sound pressure level, in decibels, at an octave band center frequency of 0 hertz.

21

$$3x + 5y = 8$$
$$9x + 15y = 24$$

For each real number r, which of the following points lies on the graph of each equation in the xy-plane for the given system?

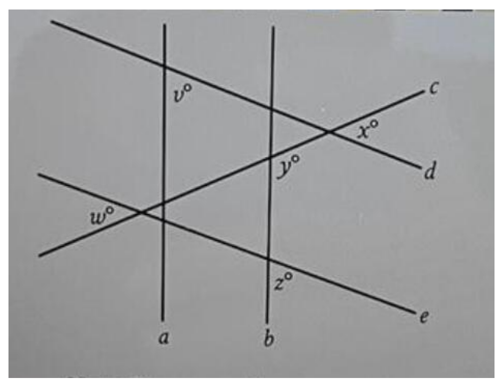
A) $(r, -\frac{5r}{3} + \frac{8}{3})$

B) $(r, \frac{3r}{5} + \frac{8}{5})$

C) $(-\frac{5r}{3} + \frac{8}{3}, r)$

D) $(\frac{r}{3} + 8, -\frac{r}{3} + 24)$

22



Note: Figure not drawn to scale.

In the figure, parallel lines a and b are intersected by lines c, d, and e. If $z = 49$, $y = 136$, and $v < z$, which statement about x and w must be true?

A) $x < w$

B) $x > w$

C) $x = w$

D) $x + w = 90$

No Test Material On This Page

	R and W 1	R and W 2	Math 1	Math 2
1	D	C	B	D
2	D	C	D	C
3	B	C	A	B
4	C	D	A	A
5	A	A	40	A
6	B	A	D	A
7	C	D	D	A
8	D	D	A	A
9	A	C	-19	19/4
10	B	C	1/3	8
11	B	D	21/250	2244
12	A	A	D	B
13	A	D	D	D
14		C	D	-11
15	D	C	343	35
16	B	B	C	6
17	C	A	C	D
18	A	B	50	A
19	D	C	B	A
20	B	B	D	A
21	A	C	C	C
22	A	A	21/29	B
23	A	C		
24	B	D		
25	D	D		
26	B	C		
27	D	A		